

BIRD MONITORING PROJECT

Easy & Natural Explanatory Video Speaking Script

Simple order: Introduction -> Cleaning -> Feature Engineering -> Dashboard -> Insights -> Conclusion

Dataset	33,251 observations
Species	78 unique species
Main tools	Python, pandas, Streamlit/dashboard
Speaking style	Question -> What I did -> What I found -> Why it matters

Use this as a speaking guide. Do not try to memorize every word. Speak naturally and use the chart title as your cue.

Slide 1 - Project Introduction

Hello everyone, my name is Fateh Mohammed Sayyed. Today I am going to explain my Bird Monitoring Project.

The main goal of this project was simple: first clean and prepare the bird data, and then use that clean data to build an interactive dashboard.

So the project has two main parts: data preparation and dashboard analysis.

The overall flow is: raw data -> cleaning -> validation -> feature creation -> dashboard -> insights.

Question: Why did you clean the data before making charts?

Easy answer: Because charts are only useful when the data behind them is correct and reliable.

Slide 2 - Data Cleaning: What I Did First

I did not directly start making charts. First, I created a clean dataset for analysis.

I loaded the data, checked the columns and data types, looked for missing values, cleaned the column names, removed empty rows and columns, fixed dates and text, removed duplicates, created time features, and then checked the final result.

The main idea is to make the data consistent and easy to use before analysis.

Question: What is the main idea of your cleaning process?

Easy answer: I changed the raw data in a controlled way so the final dataset became cleaner, more consistent, and safer to analyze.

Slide 3 - Load and Understand the Data

The first practical step was loading the CSV file with pandas.

After loading it, I checked the number of rows and columns and looked at the first few records.

This helped me understand what the original data looked like before I changed anything.

Question: Why do you inspect the data first?

Easy answer: Because I need to understand the original structure before I decide what needs to be cleaned or changed.

Slide 4 - Check Columns and Data Types

Next, I checked the column names and the data type of each column.

This is especially important for date columns. A date stored as normal text is harder to use for year, month, or weekday analysis.

So this step helped me find which columns were ready for analysis and which ones needed conversion.

Question: Why are data types important?

Easy answer: Because different operations need different data types. For example, time analysis needs real datetime values.

Slide 5 - Missing Values and Core Cleaning

After that, I checked missing values for every column and looked at the missing percentage.

I did not automatically fill or delete every missing value. First, I checked where the missing data was and how much there was.

Then I performed the main cleaning steps: I kept a copy of the original data, standardized column names, removed completely empty rows and columns, converted dates, cleaned extra spaces, treated blank text as missing, and removed duplicate records.

For time-based analysis, records without usable dates were also removed.

Question: Why did you remove duplicate records?

Easy answer: Because duplicates can increase observation counts and give the dashboard the wrong numbers.

Slide 6 - Validate the Cleaned Data

After cleaning, I checked the data again. I recalculated missing values, checked duplicate records, and looked at sample rows.

This is important because writing cleaning code is not enough. I also need to prove that the cleaning actually worked.

Question: How do you know the cleaning process worked?

Easy answer: I compare the data before and after cleaning and check the main quality indicators again.

Slide 7 - Create Time Features

Once the data was clean, I created useful time features from the date field.

I created four fields: year, month, month name, and day of the week.

This makes time-based analysis much easier because I can directly group observations by year, month, or weekday.

Question: What is feature engineering?

Easy answer: It means creating useful new variables from existing data. Here, I created time features from the original date.

Slide 8 - Dashboard Overview

Now I move from data preparation to dashboard analysis.

The dashboard contains 33,251 bird observations across 78 unique species.

I designed the dashboard around a set of simple questions: Which species are seen most often? How are birds identified? How long are the observation intervals? How is sex recorded? And how do species and identification methods connect?

Each chart answers one question, so the dashboard tells one connected story instead of showing unrelated charts.

Slide 9 - Top Species and Identification Method

First, I looked at which species have the most observations. I grouped the data by scientific name and selected the top ten species.

The most recorded species is *Cardinalis cardinalis* with 8,514 observations. It is followed by *Passerina cyanea* with 3,860 and *Spizella pusilla* with 3,542 observations.

I also checked how birds were identified. Singing is the main method, with 23,367 records, or about 70.3 percent. Calling has 6,218 records, and Visualization has 3,653.

The key insight is that a small group of species contributes a large part of the data, and singing is the main identification method.

Question: What is the value of these charts?

Easy answer: They show which species contribute most to the dataset and which identification methods are used most often.

Slide 10 - Observation Time and Sex Distribution

Next, I looked at observation interval length. The most common interval is 0 to 2.5 minutes, with 16,920 observations, or about 50.9 percent of the dataset.

Then I checked the overall sex distribution. Undetermined is the largest category with 20,269 records, around 61 percent. Male has 12,440 records and Female has 542.

So we need to be careful here. These are recorded observations, not a direct estimate of the real bird population.

Question: Can we say there are more male birds than female birds?

Easy answer: No. The dashboard shows recorded observations. Many records are undetermined, and observation effort can also affect the numbers.

Slide 11 - Species-wise Sex Comparison

Then I went one step deeper and compared Male and Female observations by species.

For example, *Passerina cyanea* has 3,700 recorded Male observations and 80 Female observations. *Cardinalis cardinalis* has 825 Male and 297 Female observations.

I also calculated a Male-to-Female ratio. For *Passerina cyanea*, the ratio is 46.25. When there are zero Female observations, the dashboard shows N/A instead of an infinite value.

The chart shows the actual observation counts, while the ratio gives a simple numerical comparison.

Question: Why use both a chart and a ratio?

Easy answer: The chart shows the size of the observations, while the ratio makes the imbalance easier to compare between species.

Slide 12 - Species Contribution and Identification Method

Finally, I connected species contribution with identification method.

I ranked all species by observation count, kept the top five as separate groups, and combined the rest into Other Species so the chart stayed easy to read.

The top five species together contribute 20,538 observations, about 61.8 percent of the full dataset. *Cardinalis cardinalis* alone contributes 8,514 observations, about 25.6 percent.

For the leading species, Singing is the main identification method, but the exact mix changes by species. *Cardinalis cardinalis* has noticeable Calling and Visualization records as well.

Question: Why group the remaining species into Other Species?

Easy answer: Because showing every species separately would make the chart crowded and harder to understand.

Slide 13 - Final Conclusion

To conclude, this project contains 33,251 bird observations covering 78 species.

The main findings are: several species make up a large part of the observations, Singing is the most common identification method, short observation intervals are common, and Undetermined is the largest sex category.

At species level, recorded Male observations are often much higher than recorded Female observations. But these numbers should not be treated as direct bird population or biological sex estimates.

The main strength of this project is the complete workflow: data preparation, cleaning, validation, feature engineering, dashboard analysis, and interpretation.

Thank you.

Quick Practice Formula - Use This for Every Chart

1. What was the question? - Tell the audience what you wanted to understand.
2. What did I do? - Briefly explain the grouping, filtering, or calculation.
3. What did I find? - Give the main insight and one useful number.
4. Why does it matter? - Explain why the result is useful for the project.

Example: “The question here was which species were observed most often. I grouped the data by scientific name and selected the top ten species. I found that *Cardinalis cardinalis* had the highest number of observations, with 8,514 records. This matters because a relatively small number of species contribute a large part of the observation volume.”